



MTP Department
SIMHADRI

Note No :1



Ref:SMTTP/MAINT. PLNG./2024-25/MTPLMI/1086954

Date:

27/03/2025(17:20:37)

Subject:Approval for revision in LMI Turbine Generator Vibration Monitoring

Sub:- Approval for Revision in LMI "Turbine Generator Vibration Monitoring" - Reg

LMI "Turbine Generator Vibration Monitoring" (LMI-COS-ISO-00-OD/OPS/MECH/007, **Rev No. - 3, Dec 2022**) was reviewed on **December 2024** and no changes were envisaged.

Further LMI-COS-ISO-00-OD/OPS/MECH/007 Rev no 04 & March 2025 is prepared based on revised COS-ISO-00-OD/OPS/MECH/007 **Rev. No.: 3, January 2025.**

The following changes were incorporated in LMI as per revised OD.

1. Clause No-1: Point - **v & vii** added.
2. Clause No-5.3: Online Vibration Monitoring System (VMS) with Analysis System.(Responsibility : C&I & **MTP* included**)
3. Clause No - 6.2, 6.2.1 **new text** added.
4. Clause No-8 "**ASSOCIATED DOCUMENTS**" added.
5. Checklist -Turbine Generator Vibration Monitoring **revised**
6. APPENDIX – B (**Responsibility-MTP-CMG**) **Point (iv)** added.
7. New Appendix - D "**Online VMAS Data Requirements**" added.

Submitted for review and further approval please

Submitted by:

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LMI is prepared as per OD/OPS/MECH//007 & May be approved

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Note No:7

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Note No:9

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Note No:10

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Note No:11

LMI is Prepared as per OD/OPS/MECH 007 & may be approved

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On:08/04/2025(18:30:31) **Ref:**SMTPP/MAINT. PLNG./2024-25/MTPLMI/1086954

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

TURBINE GENERATOR VIBRATION MONITORING

LMI-COS-ISO-00-OD/OPS/MECH/007 Rev. No.: 4

March 2025

Approved for implementation by HOP (Simhadri)

LMI APPROVAL DOCUMENT

1	LMI NAME	TURBINE GENERATOR VIBRATION MONITORING
2	LMI No.	LMI-COS-ISO-00-OD/OPS/MECH/007 Rev. No. 04
3	Reference Document	COS-ISO-00-OD/OPS/MECH/007 Rev.3, Jan 2025
4	Superseded Document	LMI-COS-ISO-00-OD/OPS/MECH/007 Rev. 03, Dec 2022
5	Prepared By	Name: Srivenkata Phani Kumar CH, Sr. Manager (MTP)
6	Checked By	Name: KVK Bhaskar, DGM (MTP) Nilanjan Mandal, AGM (TMD) Somnath Bhattacharya, AGM (Opn) Jayamol Rachel Varughese, AGM (C&I)
7	Reviewed By	Name: P.K.Jena GM (O&M)
8	Approved By	Name: Sameer Sarma HOP (Simhadri)

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TITLE	Turbine Generator Vibration Monitoring
LMI No.	LMI-COS-ISO-00-OD/OPS/MECH//007 Rev:04

1.0 Introduction

The objectives of turbine generator vibration monitoring are as follows:

- i. To indicate to the operator whether a machine is running within prescribed vibration limits as specified by the manufacturer or by internationally accepted Standards.
- ii. To detect development of faults.
- iii. To assist in diagnosing causes and identifying appropriate action when either vibration limits are exceeded or significant changes in vibration pattern are detected (Trend Analysis).
- iv. To assess and maintain a data base of machine condition on a regular basis (Monthly basis as per Appendix-C).
- v. To meet these objectives there are two general types of vibration monitoring. These are, firstly, the almost universally installed Turbine Supervisory Equipment (TSE) along with online VMAS* (Vibration monitoring & analysis system) and secondly the off-line Vibration Monitoring Equipment (VME). In most of our power stations both types of instrumentation are being used independently.
- vi. The TSE continuously measures absolute value of shaft vibration & Shell vibration and enables the operator to determine whether a unit can safely remain in service (objective (i) above), or, if there are signs of a deteriorating situation (objective (ii) above) when it may be necessary to enlist the help of specialists in carrying out further tests, measurements or analysis. The display of this information is normally in the control room and continually available to the operator.
- vii. VMAS (Vibration monitoring & analysis system) measures relative values of shaft vibration. "TSE" is provided for vibration monitoring & Protection system purpose, additionally an "Online vibration monitoring & analysis system (VMAS)" is also provided which is required for analysis purpose. For analysis of cause of high vibration, Healthy Online VMAS (Vibration Monitoring & analysis system) with availability of live & historical diagnostic data plots & trends is utmost essential for vibration analysis of High-speed rotating equipment such as steam turbine for safe operation VME (vibration monitoring equipment) is used for both offline & online vibration monitoring as described.

For Offline vibration measurement (with vibration analyser) using accelerometer for taking shell vibration by manually placing probes at point of measurement.

For online vibration measurement using 16 channel or higher analyser for taking shaft & bearing shell vibration from buffered output which will provide same data as online VMAS (in case of unavailability of online VMAS). The VMAS & VME enables an in-depth analysis of the vibration data, including phase and plant state information, and caters for high quality data to be recorded on load during run-up and run-down in order to aid diagnosis of fault conditions (objective (ii)-(iv) above). This can be accomplished

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with off-line and on-line systems, but the additional benefits of on-line systems are continuous monitoring & improved response time in the diagnosis of plant problems and better correlation of the vibration information of the state of the plant when specific problems arise. In NTPC, off-line VME system is widely used to gather reliable data for effective faults diagnosis.

2.0 SCOPE:

This document is applicable for the steam turbines and generator including exciter of the 4 X 500 MW units of SIMHADRI STPS.

3.0 Suspended Documents:

Suspended Document: LMI-COS-ISO-00-OD/OPS/MECH/007 Rev no 03 & Dec 2022.

4.0 Details of Revision

1. Clause No-1: Point - v & vii added.
2. Clause No-5.3: Online Vibration Monitoring System (VMS) with Analysis System.(Responsibility : C&I & MTP* included)
3. Clause No - 6.2, 6.2.1 new text added.
4. Clause No-8 "ASSOCIATED DOCUMENTS" added.
5. Checklist -Turbine Generator Vibration Monitoring revised
6. APPENDIX – B (Responsibility-MTP-CMG) Point (iv) added.
7. New Appendix - D "Online VMAS Data Requirements" added.

5.0 PROCEDURES:

The following are recommended procedures to enable the objectives (i to iv) to be achieved

5.1

- i. The Station Manager shall nominate a representative within the station Condition Monitoring Group who will be responsible for implementing this Directive, undertaking its routine operation and liaising with specialists as required eg. TG/SG/EG.
- ii. Turbine generators shall be operated within vibration limits as recommended by Manufactures and specified in "Operating Guidelines on Vibration limits for Turbogenerators & Rotating machines" issued by Operation Service.

(Responsibility: Operation)

- iii. Criteria shall be established defining acceptable vibration trends and step changes for each turbine generator bearing position. The appropriate vibration criteria should be displayed to the operator, and advice sought in the event of any limits being exceeded changes occurring.

(Responsibility: TMD)

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iv. A procedure shall be established defining responsibilities for operating the off-line VME and identifying arrangements for normal and emergency situations.

v. A Comprehensive record of machine build shall be maintained

(Responsibility: TMD)

- ✓ Rotor/Coupling concentricity.
- ✓ The full bearing alignment data from each machine, relative to the smooth catenary condition before and after overhauls.
- ✓ The balance weight distribution on each rotor.
- ✓ The gland clearances for each shaft.
- ✓ The angular coupling of each rotor relative to its neighbour.
- ✓ Datum block clearances to obtain the cold to hot axial movement of the pedestals and axial-transverse movement of the cylinder paws.
- ✓ Records of remedial work carried out on the turbine-generator and its support structure.
- ✓ CRO value of all journals & coupling of Turbine & Generator.
- ✓ Swing Value at HPT front & Exciter bearing.

These records should be updated each time the build of any unit is disturbed or modified.

5.2 Vibration Monitoring & Analysis

(Responsibility: MTP-CMG)

Maintain data base of vibration data & TSI parameters as per the Guidelines issued by OS for each turbine generator, in order to identify significant changes and trends, and assess generic plant implications. As per Appendix- 'C'.

5.3 Online Vibration Monitoring & Analysis System

(VMAS) (Responsibility: C&I)

Online Vibration Monitoring & Analysis system displaying diagnostic data, plots, trends is required for High-speed rotating equipment such as steam turbine for safe operation. Healthiness of online Vibration monitoring & Analysis system (Synkawa/System-1 Bentley/Meggitt/Alstom /Any other) to be ensured for analysis & detection of abnormal condition during slow roll speed, Transient speed (startup /shut down) & steady state.

Following to be ensured: -

- i. Appropriate trigger setting for capturing data for following conditions a) Slow roll speed b) Transient c) steady state condition so that any abnormal condition is not missed out.
- ii. VMAS system analytic software is having sufficient working memory & data storage capacity.
- iii. VMAS software provide all important diagnostic plots including but not limiting

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to bode plot, polar plot, full spectrum, trend plot, Shaft centreline plots, cascade plots etc.

- iv. VMAS system is having capability of data capturing & historian storage as specified in Appendix –`D`. Requirement as specified in appendix `D` represent minimum capability of system for providing data for analysis of all fault condition and thus enhance reliability of online VMAS.
- v. For data storage optimization some OEM, s has provided trigger based high resolution data capturing such as in Meggitt VMS system & others.

i.e. In case of rise in vibration to some pre-determined level, high resolution data recording is initiated. In such cases the OEM logic may be verified so that in no case fault condition data is missed out.

* Responsibility for providing point i,ii,iii,iv,v lies with C&I and Condition monitoring Engineer (MTP) needs to provide feedback to C&I for any deficiency with respect to the above clause 5.3.

6.0 Measurements

6.1 Turbine Supervisory Equipment (TSE) Continuous Measurements (Responsibility: C&I)

The TSE enables the operator to ensure that overall vibration levels are kept within prescribed limits, and to detect significant departures from normal behavior. The continuously monitoring TSE should remain the primary safeguard against faults which may result in load or excitation limits, or possibly, machine breakdown and major failure. The measured vibration amplitude should be adequately displayed to the operator together with the various agreed limits.

The performance reliability of TSE should be maintained to a high standard.

6.2 Online Vibration Monitoring & Analysis system (VMAS) / Vibration Monitoring Equipment (VME) Analysis of Measurements (Responsibility: MTP-CMG)

The online VMAS /VME enables detailed investigations to be made, and should a vibrational change be detected, provides the data to diagnose the cause.

Correct and prompt diagnosis of faults requires the systematic recording and archiving of vibrational behaviour.

Experience has shown that VMAS/VME measurements under three types of operation conditions are usually necessary to provide comprehensive data for efficient problem diagnosis, i.e.

- i. At normal load conditions
- ii. During ramp up & ramp down of turbine rolling in both hot & cold condition
- iii. On return to service following major overhauls.

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It is responsibility of condition monitoring engineer to check that all diagnostic data plots such as trend plot (vibration overall amplitude, harmonics, sub synchronous), Spectrum, Orbit plot, bode plot, polar plot, waterfall plot are available in system & recorded for review later in online VMAS system of Bentley/Alstom/Meggitt/any other make.

The TSI parameters should be documented immediately prior to the recording of off-line VME data as per Appendix – C by condition monitoring engineer at the intervals recommended in 6.2.1 and 6.2.2 of this LMI. The selection of the position and direction of the measured vibration response should be as recommended by Condition Monitoring Station Specialist.

6.2.1 Routine On-load Measurements (Responsibility: MTP-CMG)

The purpose of regular on-load measurements is to provide a continuing assessment of a machine's in-service behavior and to ensure that a detailed history is available to assist diagnosis in the event of a problem. The measurements should be taken under established normal loading and temperature conditions. Also, measurements to be taken at other loads (i.e. Part load) in addition to normal load in case of any observation such as appreciable vibration changes occurring at low loads and for ensuring a comparison with datum. It is recommended for off-line systems that the minimum increment between measurements should be analyzed weekly for known problem machine and at least monthly otherwise. Along with offline measurement, it is to be ensured by Condition monitoring engineer that online VMAS diagnostic plots are available & being recorded in historian for same load condition.

6.2.2 Routine Hot Run-down Measurements (Responsibility: MTP-CMG)

The vibration signature of a machine obtained by analyzing vibrations recorded during a hot run- down (i.e. normal working conditions) will contain much more information (for comparison with the previous hot run-down data) than can be obtained from the on-load fingerprint. This is because the run-down signature contains the response to a range of driving frequencies, rather than just 50 Hz. The measurements should be taken under established conditions and will need to cover the full range of speeds from synchronized down to 300 r.p.m. (or to less than half the first generator critical speed). It is to be ensured that hot run-down measurement diagnostic plots such as bode plot, Polar plot, shaft center line, trends, cascade waterfall etc. are recorded in online VMAS analysis system and soft copy of plots to be uploaded on MTP dept. intranet during every planned outage (or six month whichever is sooner). In case of non-availability of VMAS analytic diagnostic system with data recording capability & display in form of diagnostic plots, data capturing through buffered output (using 16 channel or better VME) and analysis is to be carried out during every planned outage (or six month whichever is sooner).

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6.2.3 Measurement on Return-to service following Major Overhaul

During the outage, rotors may have been uncoupled or exchanged, alignment altered, or the mechanical balance of individual rotors adjusted. Any of these can introduce significant changes into the dynamics of the complete system, so that the previous history of the *machine, prior to the overhaul, is no longer a reliable baseline against which to assess changes*. The objective, therefore, of making vibration measurements after a return-to service is to obtain a set of data which accurately characterizes the new status of the machine.

The types of measurements described in 6.2.1 and 6.2.2 above are required to establish a hot datum against which to assess changes. The greatest value is obtained from these measurements if they are made under the same standard conditions. To obtain hot datum values following return- to-service, the on-load measurements should be made at the earliest after reaching steady operating conditions. **The hot run-down measurements should be made at the opportunity earliest but not later than six months.**

At certain units, it may additionally be considered desirable that measurements are taken immediately after a return-to-service, before the unit has reached steady operating temperatures (e.g. where it is agreed policy for units with known high vibration levels); this may include a run- down before the machine is hot (e.g. following overspeed test) and also measurements during loading. Such measurements can only be compared with previous measurements taken under similar circumstances, and details are given in Appendix B

6.2.4 Special Investigations

If a problem is identified, the frequency and amount of detail of the monitoring may need to be increased, and additional types of measurement may be necessary, e.g. shaft vibration, pedestal vibration, etc.

7.0 RECORDS TO BE MAINTAINED

Sl No.	Record Description	Maintained by	Frequency	Remarks
1	Records must be kept for all activities as mentioned Above	Responsibility of carrying out above Activities shall be with MTP-CMG, TMD & C&I as mentioned above.	As mentioned Above.	

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8.0 ASSOCIATED DOCUMENTS

Appendix-A: Vibration measurement - general requirements

Appendix-B: Datum measurements during return-to-service from major outages

Appendix-C: Running Parameters

Appendix-D: Online VMAS Data

Requirements Checklist -Turbine Generator

Vibration Monitoring

9.0 Review:

Review Responsibility: MTP Dept.

Review Frequency: 2 yearly basis, or as and when required, whichever is lower.

Review Approval: GM(O&M) will be reviewing and HOP-Simhadri will be approving authority for any revision in this document.

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Checklist- Turbine Generator Vibration Monitoring

<u>Checklist-Turbine Generator Vibration Monitoring</u>	<u>Responsibility</u>	<u>(Complied / Not) Status</u>	<u>Remarks</u>
1.Appropriate trigger setting for capturing data for following conditions a) Slow roll speed b) Transient c) steady state condition.	C&I/ MTP*		
2. Online VMAS system is having capability of data capturing & historian storage as specified in Appendix – 'D'	C&I/ MTP*		
3.VMAS system analytic software is having sufficient working memory & data storage capacity.	C&I		
4.VMAS software provide all important diagnostic plots including but not limiting to bode plot, polar plot, full spectrum, trend plot, Shaft centerline plots, cascade plots etc.	C&I/ MTP*		
5.Checking of any logic-based trigger for high resolution data capturing.	C&I/ MTP*		
6. All TSI parameters made available in PI server.	C&I		
7. Maintain data base of vibration data & TSI parameters as per the Guidelines issued by OS for each turbine generator, in order to identify significant changes and trends, and assess generic plant implications as per Appendix-'C' weekly for known problem machine and at least monthly otherwise.	MTP		
8. Ensure all diagnostic data plots such as trend plot (vibration overall amplitude, harmonics, sub synchronous), Spectrum, Orbit plot, bode plot, polar plot, waterfall plot are available in system & stored in historian for review later in online VMAS system.	MTP		
9. The TSI parameters should be documented immediately prior to the recording of off-line VME data as per appendix – C by station condition monitoring engineer at the intervals recommended in 6.2.1 and 6.2.2 of this LMI	MTP		
10. Along with offline measurement, it is to be ensured by Condition monitoring engineer that online VMAS diagnostic plots are available & being recorded in historian for same load condition.	MTP		

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11.1 As stated in 6.2.2. normal Routine Hot Run- down Measurements: During every planned outage or six month which is sooner, to be ensured all diagnostic plots such as bode plot, Polar plot, shaft center line, trends, cascade waterfall etc. are recorded in online VMAS analysis system and soft copy of plots to be uploaded on MTP dept.	MTP		
11.2 In case of non-availability of VMAS analytic diagnostic system with data recording capability & display in form of diagnostic plots, data capturing through buffered output (using 16 channel or better VME) and analysis is to be carried out during every planned outage (or six month whichever is sooner).	MTP		
12. Measurement on Return-to service following Major Overhaul: To obtain hot datum values following return-to-service, the on-load measurements should be made at the earliest after reaching steady operating conditions. The hot run-down measurements should be made at the opportunity earliest but not later than six months.	MTP		
13. Return-to-Service from Major Outages i) Vibration measurements should be taken during the run-up to full speed, during an overspeed test and when unit is synchronized. For transient state when speed is changing such as rolling, coasting, overspeed, it is to ensured data is being captured in online VMAS system. If online VMAS is not available, by means portable VME such as 16 channel or higher analyzer vibration to be recorded and keep in soft format for future references.	MTP		
13. ii) Vibration should be logged during the loading pattern. This may be carried out continuously or intermittently, depending on the anticipated progress in loading. It is extremely useful to have readings of overall levels (1/rev, 2/rev magnitude and phase) at 'no load', approximately 75 per cent and 100 per cent stabilized full load. If there is clear evidence of a generator 'thermal' effect, then additional interim load measurements should be advised.	MTP		
14. After every overhauling shaft reference gap voltage for all bearings set at zero speed to be kept in record. Also shaft gap voltage when machine is put in barring, at soaking, at rated speed and at 50%, 75% and 100% load to be kept in record.	MTP		

* "Responsibility for point 1,2,4,5 lies with C&I and Condition monitoring Engineer (MTP) needs to provide feedback to C&I for any deficiency with respect to the above clause 5.3".

TITLE	Turbine Generator Vibration Monitoring
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APPENDIX - A

VIBRATION MEASUREMENT - GENERAL REQUIREMENTS

Responsibility: HOD (C&I)

1. In order to provide reliable vibration data from turbine generators in a form suitable for software based computational analysis and comparison with control room displays. All TSI parameters made available in PI server.
It is recommended that the following measurements and signals are made available.
2. Signal Requirements
The following signals should be made available:
 - i. A raw (i.e. unfiltered) vibration signals for each bearing
 - ii. A Key Phaser signal

Vibration signal conditioning equipment should provide an A.C. unfiltered signal proportional to the vibration level and be able, on demand, to transmit a calibration signal of known frequency.
3. X & Y direction (orthogonal) shaft vibration probe along with shell vibration probe to be installed in all bearings of turbine, Generator & Exciter and Key phaser probe at bearing no.2 (In IP collar) if not available. Key phaser signal & vibration signals of each bearing must be available at VMS panel in UCB.
4. Turbine Trip protection to be incorporated in protection ckt as per 2 out of 14 (500MW KWU Unit) or 2 out of 10 or 12 (for 200/210/250MW KWU unit as applicable) probes shaft vibration signal logic, i.e if any two probes shaft vibration values goes beyond trip level, turbine trip signals to be initiated. In old machines, where only single vibration probe is available (X-dir), the same trip protection also to be incorporated as per two probes vibration signal logic.
Operating guidelines of shaft & pedestal (shell) absolute vibration of all turbo generator sets (200MW, 210 MW, 250MW & 500 MW KWU UNITS) are as follows:
 - ✓ Alarm Level of Pedestal Vibration: 84 microns (Pk-Pk)
 - ✓ Trip Level of Pedestal vibration: 104 microns (Pk-Pk).
 - ✓ Alarm level of Shaft vibration level: 200 microns (Pk-Pk)
 - ✓ Trip Level of Shaft Vibration Level: 300 microns (Pk-Pk).

A time delay of one second for tripping on vibration values is to be kept.

In addition to the above, if any bearing pedestal actual vibration crosses the trip value, then the machine shall be hand tripped.

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5. A Phase-locked Once/Rev Signal

It is essential that the above signal is maintained to a high standard, **as loss of this signal will preclude the accurate analysis and comparison with earlier data, necessary for diagnosis of the turbine generator condition. The phase signal, which provides basic information and further updated details can be obtained from local Condition Monitoring Group at sites.**

It is essential that, whichever system is used, the ***shaft reference is neither lost during operation of the turbine generator (due to axial expansion of the rotors),*** nor while on an outage for overhaul. Dual Tachometer signals are advised and to prevent total loss under extreme fault conditions, they should be fitted (Reflecting tape) at either end of the machine in exposed portion of the shaft aligned with the permanent key phaser probe signal.

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APPENDIX-B

DATUM MEASUREMENTS DURING RETURN-TO-SERVICE FROM MAJOR OUTAGES

Responsibility-MTP(CMG)

Under certain circumstances it may be considered necessary to take datum measurements on plant during return-to-service, while the plant is still reaching its working temperatures. The existence of this data will enable detailed comparisons to be made between the new vibration response and that from a previous return-to-service. A detailed knowledge of any changes in response will often enable a precise diagnosis to be made swiftly on the basis of changes to the build state and known defects.

- i) Vibration measurements should be taken during the run-up to speed, during an over speed test and when unit is synchronized.
- ii) Vibration should be logged during the loading pattern. This may be carried out continuously or intermittently, depending on the anticipated progress in loading.

It is extremely useful to have readings of overall levels (1/rev, 2/rev magnitude and phase) at 'no load', approximately 75 per cent and 100 per cent stabilized full load. If there is clear evidence of a generator 'thermal' effect, then additional interim load measurements should be advised.

- iii) In addition, it may be considered beneficial to arrange for a hot run-down from full speed to approximately 300 rev/min and measure the vibration signatures throughout this range, at the earliest opportunity following a return-to-service and prior to the first natural 'hot-run-down'. The analysis of this reading then forms the as-built 'hot run- down' datum. Alternatively, a warm (or cold) run-down could be obtained for base load plant, during the return-to-service. MTP CMG to ensure following transient data is captured in online VMAS system and take soft copy of plots (bode plot, polar plot, shaft center line plot, any additional plots such as full spectrum cascade etc. on case to case basis as per condition monitoring analyst and maintain it for ready reference on intranet as reference datum.
- iv) After every TG overhauling, shaft reference gap voltage for all bearings at zero speed to be kept in record. Also shaft gap voltage when machine is put in barring, at soaking speed
, at rated speed and at 50% load , 75% load and 100% load to be kept in record.

In considering such datum measurements, it is worth recognizing the complex nature of the vibration response of rigidly coupled multi-rotor machines. The vibrations depend not only on residual defects in the mechanical build of the machine, but also on factors controlling its operation, e.g. steam conditions, electrical generation, and bearing oil supply conditions. In all cases, the vibration response is determined both by the dynamic exciting forces present and by the machine's response to these forces. The excitation forces that can arise from defects in the mechanical build

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including those due to residual unbalance and lack of straightness of individual rotors, together with any bends which result when the rotors are coupled. The response to these excitation forces is considerably influenced by the dynamics of the journal bearings and depends on the bearing loadings which are a function of bearing alignment.

The process of electrically loading the machine induces 'thermal' changes which modify both the dynamic excitation forces and the machine's response to these forces. The excitation forces are modified because of thermally induced bending of rotors (in particular the generator rotor). Relative movements of bearings during operation result in bearing load changes which, in turn, modify the response to a given unbalance.

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APPENDIX –C: Running Parameters Responsibility: MTP(CMG)

S No.	Parameter	Measurement unit	UNIT No-				
1	Date						
2	Time						
3	Load	MW					
4	Frequency	Hz					
5	Mvar	Mvar					
6	Gen Rotor Voltage	KV					
7	Gen Field Current	Amp					
8	Gen Exciter Current	Amp					
9	MS Press before turbine	Kg/Cm ²					
10	1 st Stage Pressure	Kg/Cm ²					
11	MS Temp before turbine(L/R)	° C					
12	HRH Press at IPT inlet	Kg/Cm ²					
13	HRH Temp at IPT inlet(L/R)	° C					
14	Condenser Vacuum	Kg/Cm ²					
TG Pedestal Vibration			UCB Online Pedestal Vibration		Offline Pedestal		
S No.	Parameter	Measurement unit	X Direction	Y direction	Hor	Ver	Axial
15	Bearing – 1	Micron Pk-Pk					
16	Bearing - 2	Micron Pk-Pk					
17	Bearing - 3	Micron Pk-Pk					
18	Bearing - 4	Micron Pk-Pk					
19	Bearing - 5	Micron Pk-Pk					
20	Bearing - 6	Micron Pk-Pk					
21	Bearing - 7	Micron Pk-Pk					
22	Bearing - 8	Micron Pk-Pk					
23	Bearing - 9	Micron Pk-Pk					

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24	Bearing - 10	Micron Pk-Pk								
TG Shaft Absolute Vibration (UCB-DCS value)			X Probe				Y Probe			
25	Bearing – 1	Micron Pk-Pk								
26	Bearing - 2	Micron Pk-Pk								
27	Bearing - 3	Micron Pk-Pk								
28	Bearing - 4	Micron Pk-Pk								
TG Shaft Absolute Vibration (UCB-DCS value)			X Probe				Y Probe			
29	Bearing - 5	Micron Pk-Pk								
30	Bearing - 6	Micron Pk-Pk								
31	Bearing - 7	Micron Pk-Pk								
32	Bearing - 8	Micron Pk-Pk								
33	Bearing - 9	Micron Pk-Pk								
34	Bearing - 10	Micron Pk-Pk								
TG Relative Shaft Vibration [From Online VMS system]		Micron Pk-Pk	X Probe				Y Probe			
			OA	1X	Phase	Gap Volt	OA	1X	Phase	Gap Vol
35	Bearing - 1	Micron Pk-Pk								
36	Bearing - 2	Micron Pk-Pk								
37	Bearing - 3	Micron Pk-Pk								
38	Bearing - 4	Micron Pk-Pk								
39	Bearing - 5	Micron Pk-Pk								
40	Bearing - 6	Micron Pk-Pk								
41	Bearing - 7	Micron Pk-Pk								
42	Bearing - 8	Micron Pk-Pk								
43	Bearing - 9	Micron Pk-Pk								
44	Bearing - 10	Micron Pk-Pk								

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S No.	Parameter	Measurement unit	UNIT No-
45	CEP Suction Hdr Temp	° C	
46	CW inlet Temp L/R	° C	
47	CW outlet Temp L/R	° C	
48	CW Flow	T/Hr	
49	Seal Steam Hdr Temp	° C	
50	ACC value	μS/Cm	
51	Seal Steam Hdr Press	Mmwcl	
52	TG lube oil Press	Kg/Cm ²	
53	TG lube oil Temp after cooler/Valve position	° C	
54	Shaft Expansion - HPT-F	Mm	
55	Shaft Expansion - IPT-R	Mm	
56	LPT (R) - Diff. Expansion	Mm	
57	HP - Diff Expansion	Mm	
58	IP - Diff Expansion	Mm	
59	Over all Expansion - HP	Mm	
60	Over all Expansion - IP	Mm	
61	Axial Shift-1	Microns	
62	Axial Shift-2	Microns	
63	Axial Shift-3	Microns	
64	TG Bearing -01	TL/T R ° C	
65	TG Bearing -01	BL/B R ° C	
66	TG Bearing -02	TL/T R ° C	
67	TG Bearing -02	BL/B R ° C	
68	TG Bearing -03	BF/B R ° C	
69	TG Bearing -04	BF/B R ° C	

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70	TG Bearing -05	° C		
71	TG Bearing -06	° C		
72	TG Bearing -07	° C		
73	TG Bearing -08	° C		
74	TG Bearing -09	° C		
75	TG Bearing -10	° C		
76	Thrust Brg. Temp (Working Side)	° C		
77	Thrust Brg. Temp (Non-Working Side)	° C		
78	TG Brg-1 Return Oil Temp	° C		
79	TG Brg-2 Return Oil Temp	° C		
80	TG Brg-3 Return Oil Temp	° C		
81	TG Brg-4 Return Oil Temp	° C		
82	TG Brg-5 Return Oil Temp	° C		
83	TG Brg-6 Return Oil Temp	° C		
84	TG Brg-7 Return Oil Temp	° C		
85	TG Brg-8 Return Oil Temp	° C		
86	TG Brg-9 Return Oil Temp	° C		
87	TG Brg-10 Return Oil Temp	° C		

The above data shall be shared with CC-OS on monthly basis.

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**APPENDIX - D: Online VMAS Data Requirements
(Responsibility-C&I and MTP-CMG)**

A. Transient data (During Rolling & coasting down)

1. High resolution data capture in transient state i.e. All startup 100 RPM to 2950 RPM + 20 minute & all shutdown from 2950 RPM to 100RPM +20 min.

2. Data saving interval during transient: Trend 1 sec interval

(Trend of Rotation speed, OA amplitude, GAP voltage, 0.5X amplitude / phase, 1X amplitude / phase, 2X amplitude / phase, Not-1X amplitude, nX1 to nX4 amplitude/phase)

Also, polar plot, bode plot which takes input as 1x amplitude & phase, & shaft centre line derived from gap voltage values is also required in 1 sec interval.)

Waveform 10 sec interval: Time waveform, Full Spectrum, Orbit plot during transient should be at least captured in 10 sec intervals.

3. All transient events (startup & shutdown) of at least 2 years should in available in software database.

B. Steady state data -short term (Preceding 30 days data from present)

1. Trend 10 sec interval

(Trend of Rotation speed, OA amplitude, GAP voltage, 0.5X amplitude / phase, 1X amplitude / phase, 2X amplitude / phase, Not-1X amplitude, nX1 to nX4 amplitude/phase)

Also, polar plot, bode plot which takes input as 1x amplitude & phase, & shaft centre line derived from gap voltage values is also required in 10 sec intervals.)

2. Waveform 10 sec interval: Time waveform, Full Spectrum, Orbit plot during transient should be at least captured in 10 sec intervals.

C. Steady state High resolution data Trigger by alarm

1. High resolution data 3 hours of data before and 3 hours of data after the alarm occurred.

Trend 1 sec interval (Trend of Rotation speed, OA amplitude, GAP voltage, 0.5X amplitude / phase, 1X amplitude / phase, 2X amplitude / phase, Not-1X amplitude, nX1 to nX4 amplitude/phase) Also, polar plot, bode plot which takes input as 1x amplitude & phase, & shaft centre line derived from gap voltage values is also required in 1 sec intervals.

2. Waveform 10 sec interval: Time waveform, Full Spectrum, Orbit plot during transient should be at least captured in 10 sec intervals.

3. Alarm triggered high resolution data should of at least 2 years should in available in software database.

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D. Steady state data -long term (2 years data)**1. Trend 10 min interval**

(Trend of Rotation speed, OA amplitude, GAP voltage, 0.5X amplitude / phase, 1X amplitude / phase, 2X amplitude / phase, Not-1X amplitude, nX1 to nX4 amplitude/phase)

Also, polar plot, bode plot which takes input as 1x amplitude & phase, & shaft Centre line derived from gap voltage values is also required in 10 sec intervals.)

2. Waveform 10 min interval:

Time waveform, Full Spectrum, Orbit plot during transient should be at least captured in 10 sec intervals.

*Responsibility for Annexure-D lies primarily with C&I but Condition monitoring Engineer (MTP dept.) needs to verify and confirm the requirement laid as per Annexure D are fulfilled or not by VMAS system and give feedback to C&I. In case of system deficiency, the same should be brought to notice to station C&I by MTP dept.